Career Analysis of Lionel Messi : An International Data Study (1872–2026)

Exploratory Data Analysis using Python, Pandas, and Seaborn.

The Goal is to analyse the scoring patterns and career longevity of one of football's most renowned players using a historical dataset of 49,000+ matches.

Lionel Messi

Career Span: 2005 – Present

The Goal: To understand how a young winger transformed into the world's most consistent goal-scoring leader.

Data Cleaning and Processing

This project includes 4 different datasets(results, shootouts, goalscorers and former names).

The Processing Pipeline

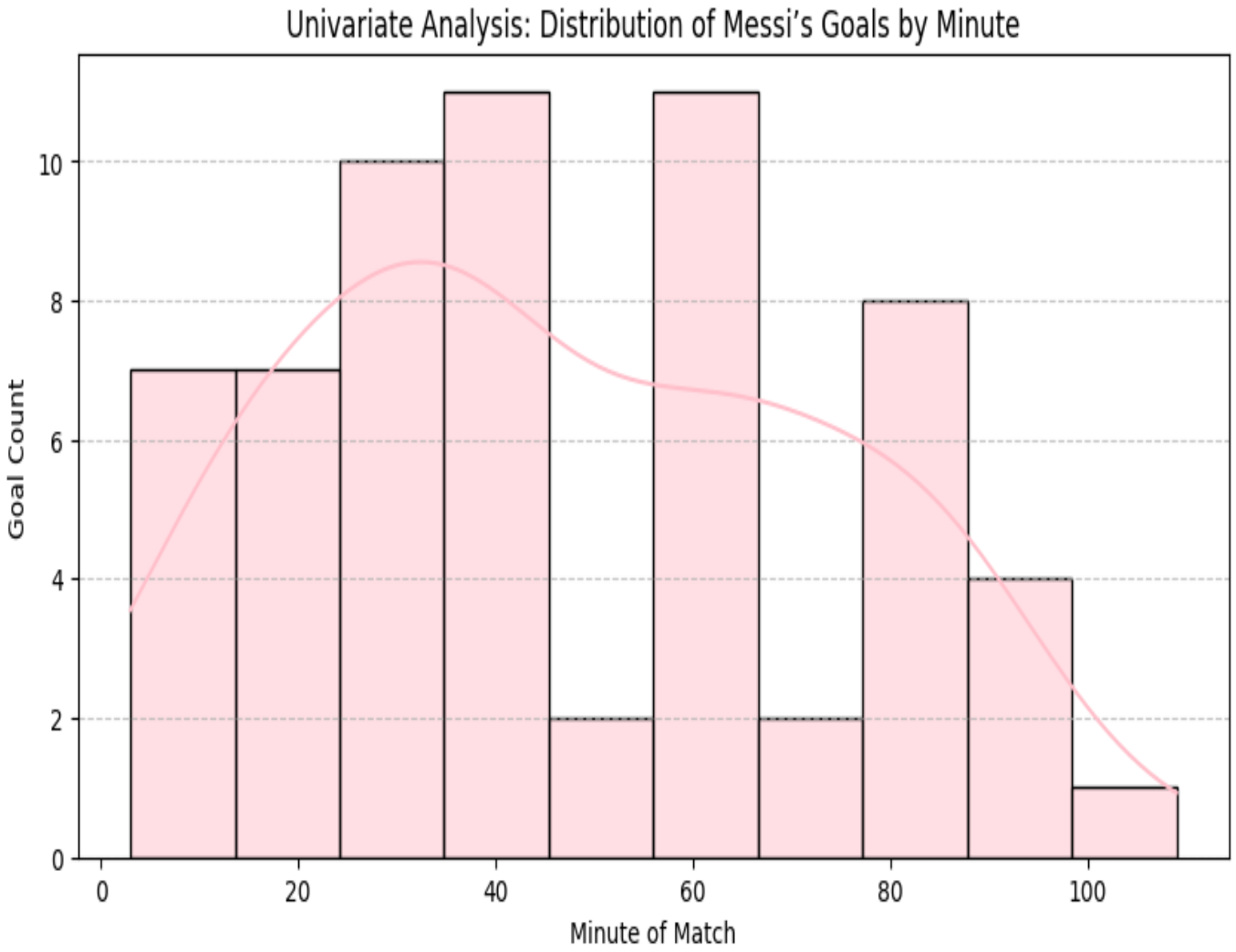
> Converting dates into a uniform datetime format to allow for accurate time-series analysis.

>Merging the goalscorers and results files to attach "Tournament Context" to individual goals.

>Ensuring historical names (e.g., Soviet Union) were mapped to modern equivalents for consistent grouping.

>Using tools like Python(for data manipulation and cleaning) and matplotlib & Seaborn (for Visuals narratives).

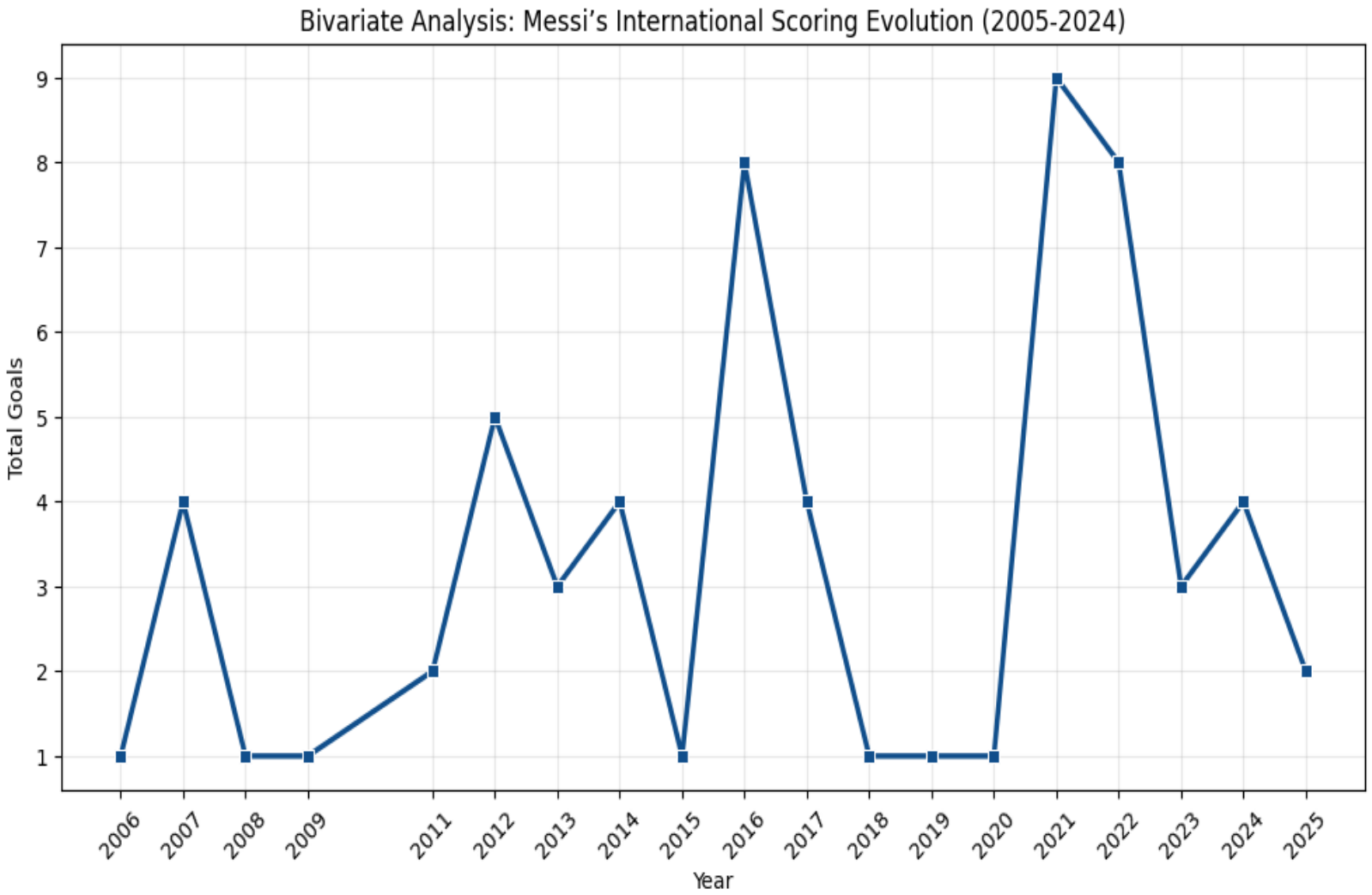
When Does the Magic Happen? (Univariate Analysis)



This indicates "High Pressure Success." Messi excels when the game is tightest and the clock is running out. He doesn't just score; he scores when it matters most.

The Finding: Unlike many strikers who score early, Messi has a high concentration of goals in the **final minutes of each half** (the 40-45' and 85-90' marks).

The Transformatory Period (Bivariate Analysis)



2012 was the year Messi broke the world record for most goals in a calendar year (91). The 2021-2023 peak aligns victories, showing he became *more* efficient as he got older.

The Finding: There is a massive "jump" in his scoring output starting in **2012**, and another sustained peak from **2021 to 2023**.

Best vs. Worst Years

The Peak (2022): His most impactful year, where his goal-scoring led directly to a World Cup trophy.
The "Quiet" Years (2005-2006): These were his learning years. The data shows low scoring but high participation, proving that greatness takes time to build.
Feature Evolution: His early career involved more "open play" goals, while his later career shows an increase in set-piece and penalty efficiency.

Conclusion

Messi’s career isn't just a straight line; it’s a series of "evolutions." He adapted his game as his speed changed, ensuring he remained the most dangerous player on the pitch for nearly two decades. Local Argentine media often cited 2011/2012 as the 'Transformation of the Captain,' where Messi fully embraced the leadership role, which is reflected in the sharp upward trend in the Bivariate Analysis plot."

THANK YOU